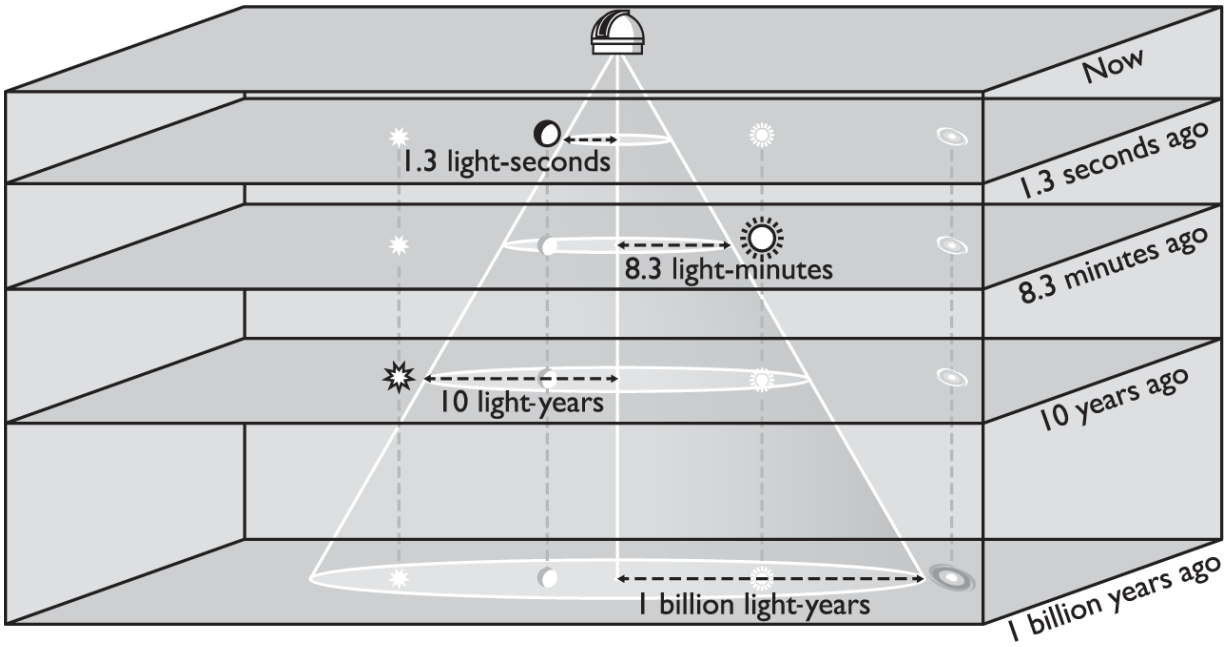


Figure 2: Diagram of light moving through spacetime. In this diagram, time moves forward in the upward direction, and we're showing only two dimensions of space instead of all three. The positions of four objects that are stationary in space are represented by the vertical dotted lines, marking the same location at different times. The "light cone" is the region we can see in the past from the observatory—it encompasses everything close enough to us that the light has had time to reach us since it was emitted. We can see a galaxy a billion light-years away as it was a billion years ago, but we can't see what it looks like "now," because the "now" version of that galaxy is outside our light cone.

But what if we look back even farther? What if we look back to the time before stars, before galaxies, before hydrogen? Can we see the Big Bang itself?

Yes. We can.

SEEING THE BIG BANG

There's a popular picture of the Big Bang as some kind of explosion—a sudden conflagration of light and matter from a single point that billowed out through the universe. It wasn't like that. The Big Bang wasn't an explosion within the universe, it was an expansion *of* the universe. And it didn't happen at a single point, but at *every* point. Every point in space in the universe today—a spot on the edge of a distant galaxy, a piece of intergalactic space just as far in the other direction, the room in which you were born—every one of these points was, at the beginning of time, close